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PROJECT PROPOSAL TOPICS

1. Smart Actuated Valve with Predictive Maintenance

This project aims to design and develop an intelligent actuated valve system capable of performing both automatic flow control and condition monitoring. Unlike conventional actuated valves that only open or close in response to control signals, the proposed system will monitor key operating parameters such as actuator current, valve position, vibration, and temperature to detect signs of wear, sticking, or mechanical failure before a breakdown occurs. The project combines mechanical design, sensors, embedded control, and predictive maintenance techniques to improve equipment reliability, reduce maintenance costs, minimize unplanned downtime, and extend the service life of industrial valve systems used in water distribution, manufacturing, and oil and gas industries.

2. Self-Balancing Two-Wheel Robot

This project aims to design and construct a two-wheel self-balancing robot capable of maintaining stability using feedback control. The robot will use sensors such as an inertial measurement unit (IMU) to continuously detect its tilt angle while a control algorithm adjusts the speed of the motors to maintain balance. Unlike many existing educational balancing robots that are tested only on smooth surfaces, this project will investigate the robot's stability and control performance on uneven terrain. The project applies principles of dynamics, control systems, mechatronics, and robotics to improve balancing performance while maintaining a simple and low-cost design suitable for educational and research applications.

3. Self-Balancing Cube

This project aims to develop a self-balancing cube capable of balancing on one edge or corner using internally mounted reaction wheels. The balancing process is achieved by controlling the rotational speed of the reaction wheels, generating the necessary torque to counteract the cube's motion and maintain equilibrium. The project investigates the design and implementation of a low-cost balancing system using readily available electronic components while improving balance duration, disturbance rejection, and overall stability. It combines mechanical engineering, dynamics, control systems, embedded programming, and robotics, providing a practical platform for studying advanced control techniques used in aerospace and robotic applications.

4. CubeSat Deployable Solar Panel Mechanism

This project aims to design and develop a lightweight deployable solar panel mechanism for CubeSat satellites. The mechanism will securely store the solar panels during launch and automatically deploy them once the satellite reaches orbit, ensuring maximum power generation while maintaining structural reliability. The project focuses on developing a low-cost deployment mechanism suitable for educational CubeSats by investigating lightweight materials, hinge mechanisms, locking systems, and deployment reliability. The outcome will contribute to improving the accessibility of CubeSat development for universities and research institutions while reducing overall system complexity and manufacturing cost.

5. Satellite Robotic Arm Prototype

This project aims to design and fabricate a lightweight robotic arm prototype capable of performing simple satellite servicing operations such as object manipulation, docking assistance, and inspection tasks. The robotic arm will be designed using mechanical linkages, servo actuators, and embedded control systems to simulate robotic operations commonly performed in space missions. The project addresses the need for affordable educational research platforms by developing a low-cost prototype that demonstrates the principles of robotic manipulation, motion control, and spacecraft servicing without the high cost associated with commercial space robotics.

6. CubeSat Propulsion Module

This project aims to develop a miniature propulsion module suitable for CubeSat satellites to perform orbital corrections and small maneuvering operations. The study will investigate compact propulsion concepts that can be safely demonstrated within a laboratory environment while emphasizing lightweight construction, modularity, and ease of integration with educational CubeSat platforms. The project seeks to address the shortage of affordable propulsion systems available to universities by developing a simplified propulsion module that can support research in spacecraft dynamics, mission planning, and orbital maneuvering.

7. Space Debris Capture Mechanism

This project aims to design and evaluate a mechanical capture mechanism capable of retrieving simulated space debris under laboratory conditions. Various capture concepts, including claw, net, harpoon, and soft robotic mechanisms, will be investigated to determine their effectiveness in securing objects of different shapes and sizes. The project focuses on improving capture efficiency, reliability, and mechanical simplicity while contributing to ongoing research on active space debris removal. The resulting prototype will provide a low-cost experimental platform for studying debris capture technologies applicable to future satellite servicing and orbital cleanup missions.

8. Modular Rocket Engine for University Research

This project aims to design a modular laboratory-scale rocket engine platform that allows different engine components, such as the nozzle, injector, combustion chamber, and fuel feed system, to be easily replaced and tested. Unlike conventional educational rocket engines that are built as fixed systems, the proposed design will enable researchers to experimentally compare the performance of different configurations using a single test platform. The project

will contribute to propulsion research by providing an affordable experimental facility that supports repeated testing, design optimization, and engineering education while reducing development cost and fabrication time.

9. Orbital Debris Capture Test Facility

This project aims to design and construct a laboratory test facility for evaluating space debris capture mechanisms under controlled conditions. Rather than developing a spacecraft, the project focuses on building experimental equipment capable of testing different capture devices, measuring capture efficiency, and analyzing the mechanical behavior of simulated debris interactions. The proposed facility will provide universities and research laboratories with a reusable platform for conducting experiments related to orbital servicing, robotic capture systems, and space debris mitigation while reducing the cost of conducting aerospace research.

10. Satellite Docking Mechanism Prototype

This project aims to develop a compact mechanical docking mechanism suitable for small satellites such as CubeSats. The docking mechanism will be designed to enable secure alignment, capture, and connection between two satellite models during simulated docking operations. The project investigates different mechanical locking mechanisms, alignment features, and structural configurations to improve docking reliability while maintaining a lightweight and low-cost design. The resulting prototype will serve as an educational research platform for studying spacecraft docking technologies and may contribute to future developments in satellite servicing, formation flying, and in-orbit assembly.